

## Cosine rule



1  $\cos \theta = \frac{b^2 + c^2 - a^2}{2bc}$

2

a  $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$

b  $a^2 = b^2 + c^2 - 2bc \cos A$

c  $b = \sqrt{a^2 + c^2 - 2ac \cos B}$

3

a  $\cos Q = \frac{p^2 + r^2 - q^2}{2pr}$

b  $r^2 = p^2 + q^2 - 2pq \cos R$

4  $\angle ABC$

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## Unknown sides

5

a  $a = 14.14$

b  $a = 212.34$

c  $q = 19.23$

d  $c = 10.60$

e  $a = 14.11$

f  $c = 13.17$

g  $y = 6.52$

h  $y = 49.40$

i  $z = 21.67$

j  $w = 12.75$

6  $AB = 3\sqrt{2}$

7  $v = 5$

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## Unknown angles

8

a  $w = 42^\circ$

b  $w = 32^\circ$

c  $A = 27^\circ$

d  $A = 34^\circ$

e  $A = 39^\circ$

f  $B = 51^\circ$

g  $B = 53^\circ$

h  $C = 104^\circ$

9  $\theta = 48.24^\circ$

10  $B = 41^\circ 47' 4''$

11  $\cos Q = -\frac{5}{11}$

12  $\theta = 24^\circ$



13  $x = 142^\circ$

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## Applications

14  $x = 34.78 \text{ km}$

15  $x = 40^\circ$

16  $x = 48.57$

17  $x = 45.7 \text{ cm}$

18  
a  $x = 277.3 \text{ km}$

b  $568.2 \text{ km}$

19  $x = 53.6 \text{ m}$